STATISTICS

Chapter 3 Homework

Probability and conditional reasoning

Sections 3.1-3.5 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

A four-function calculator is sufficient; needed advanced numerical values are supplied. Show probability setups, units, and brief explanations. Round probabilities to four decimals unless exact fractions are convenient. All scenarios and data are fictional.

Events and denominators

1. Roll a fair six-sided die. List the sample space. Let A be an even result and B a result greater than 4. List A, B, their intersection, their union, and the complement of A; find the probability of each listed event.

2. For the die events in Problem 1, are A and B mutually exclusive? Are they independent? Use a numerical test for each claim.

3. A student is chosen uniformly from this table of 100 students. Compute the probability of using tutoring, attending morning class, and both. Add all missing row and column totals.

	Tutoring	No tutoring
Morning	30	20
Afternoon	10	40

4. Using Problem 3, find P(tutoring | morning) and P(morning | tutoring). State the denominator group in each case and explain why the answers differ.

5. Using Problem 3, find P(morning OR tutoring). Are these events independent? Support your answers with calculations.

Trees, complements, and dependence

6. A bag holds 3 red and 2 blue tokens. Draw two tokens uniformly without replacement. Draw a probability tree for the two colors and label every branch.

7. Using the tree in Problem 6, find the probability of (a) two red tokens, (b) exactly one red token, and (c) at least one red token. Show how you combine paths or use a complement.

8. Repeat only the probability of two red tokens if the first token is replaced before the second draw. Explain what changed about independence.

9. Each of three independent deliveries arrives on time with probability 0.8. Find the probability that at least one is late. Explain why 3×0.2 is not the answer.

10. Out of 60 people, 25 use service A, 20 use service B, and 8 use both. Draw a Venn diagram with counts in all four regions. Find the probability that a randomly chosen person uses neither.

11. A fair coin lands heads five times. Is tails more likely on the next toss? Explain in terms of the model assumptions.